

2.2.3.1.3 Cleaning the *Sbytes* and *Dur* column()

```

# Libraries
library(tidyverse)
library(skimmr)
library(ggplot2)
library(corrplot)
library(dplyr)

# Load dataset
df <- read.csv("/Users/moqui/Downloads/UNSW-NB15_uncleaned.csv")
# Reading characters
df_all <- readr::read_csv("/Users/moqui/Downloads/UNSW-NB15_uncleaned.csv",
  col_types = readr::cols(.default = "c"))
view(df)

# --- Convert to numeric (remove "." or other symbols) ---
df$sbytes <- as.numeric(gsub("[^0-9\\.]", "", df$sbytes))
df$dur <- as.numeric(gsub("[^0-9\\.]", "", df$dur))

# Distribution before cleaning
ggplot(df) + geom_boxplot(aes(y = sbytes), fill = "lightblue") + ggtitle("Distribution of sbytes pre cleaning")
ggplot(df) + geom_boxplot(aes(y = dur), fill = "lightpink") + ggtitle("Distribution of dur pre cleaning")

# Check missing values
sum(is.na(df$sbytes))
sum(is.na(df$dur))

# --- Remove impossible (negative) values ---
df$sbytes[df$sbytes < 0] <- NA
df$dur[df$dur < 0] <- NA

# --- Detect outliers using IQR Method for sbytes ---
Q1_sb <- quantile(df$sbytes, 0.25, na.rm = TRUE)
Q3_sb <- quantile(df$sbytes, 0.75, na.rm = TRUE)
IQR_sb <- Q3_sb - Q1_sb

lower_sb <- Q1_sb - 1.5 * IQR_sb
upper_sb <- Q3_sb + 1.5 * IQR_sb

df$sbytes[df$sbytes < lower_sb | df$sbytes > upper_sb] <- NA

# --- Detect outliers using IQR Method for dur ---
Q1_dur <- quantile(df$dur, 0.25, na.rm = TRUE)
Q3_dur <- quantile(df$dur, 0.75, na.rm = TRUE)
IQR_dur <- Q3_dur - Q1_dur

lower_dur <- Q1_dur - 1.5 * IQR_dur
upper_dur <- Q3_dur + 1.5 * IQR_dur

df$dur[df$dur < lower_dur | df$dur > upper_dur] <- NA

# --- Median Imputation for both variables ---
median_sbytes <- median(df$sbytes, na.rm = TRUE)
median_dur <- median(df$dur, na.rm = TRUE)

df$sbytes[is.na(df$sbytes)] <- median_sbytes
df$dur[is.na(df$dur)] <- median_dur

# --- Log Transformation (recommended for skewed variables) ---
df <- df %>%
  mutate(
    log_sbytes = log10(sbytes + 1),
    log_dur = log10(dur + 1)
  )

# Summary after cleaning
#
summary(df$sbytes)
summary(df$dur)
summary(df$log_sbytes)
summary(df$log_dur)

summary(df)

```

```

# Visuals after cleaning

# making df a dataframe
df <- as.data.frame(df)
# creating structure and format
df_ref <- df %>%
  dplyr::select(sbytes, dur) %>%
  tidyr::pivot_longer(
    cols = c(sbytes, dur),
    names_to = "variable",
    values_to = "value"
  )

# Determining 5th - 95th percentile
lower_lim <- quantile(df_ref$value, 0.05, na.rm = TRUE)
upper_lim <- quantile(df_ref$value, 0.95, na.rm = TRUE)

#Boxplot after cleaning sbytes and dur
ggplot(df_ref, aes(x = variable, y = value, fill = variable, color = variable)) +
  geom_boxplot(outlier.alpha = 0.2, width = 0.6) +
  coord_cartesian(ylim = c(lower_lim, upper_lim)) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Boxplots of cleaned sbytes and dur",
    x = "variable",
    y = "value"
  ) +
  theme(legend.position = "none")

```

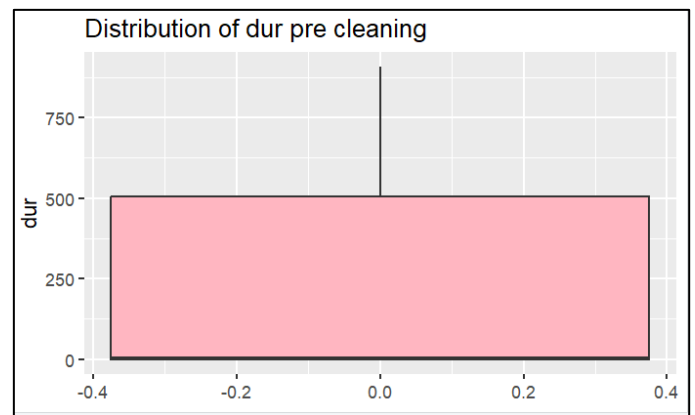
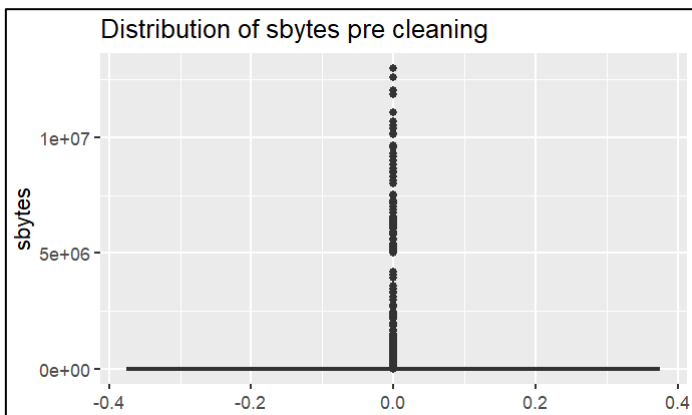


Figure 18 – Boxplot shown before cleaning of dur and sbyte

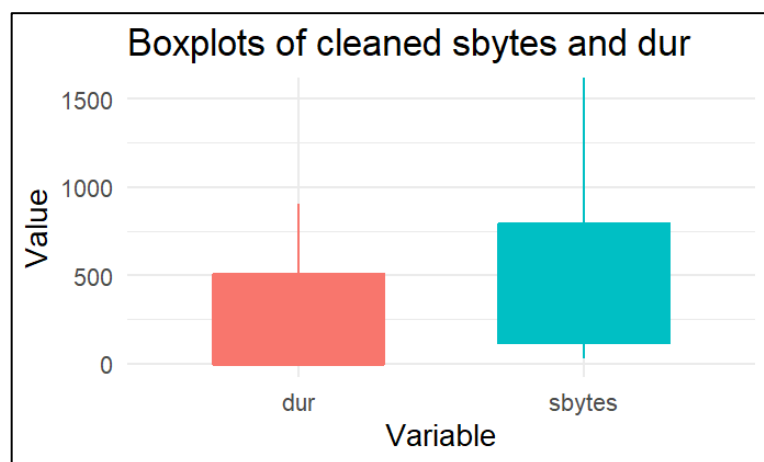


Figure 19 – Boxplot shown after cleaning of dur and sbyte

3.3 Objective 3 : Assessing the relationship between Sbytes and Dur towards the presence of attack. - (Nashrah Fannana Moquit -)

Analysis 3.3.1:How do the distributions of log(sbytes) and log(dur) differ between Normal and Attack?

```
#Attack indicator
df$attack_flag <- ifelse(df$attack_cat == "Normal", 0, 1)

df_clean <- df %>% drop_na()
cat("After non-negative filter rows:", nrow(df_num), "\n")

#log 1p
df_num <- df_num %>%
  mutate(
    log_sbytes = log1p(sbytes),
    log_dur = log1p(dur)
  )
```

Figure 44– Codes for creating attack indicator and log1p transformation

The initial codes of data loading and cleaning has validated the raw network traffic dataset and prepared it for analysis. Identifying absent values, inconsistent formatting, and added underscores aids in reducing structural errors, while transforming dur and sbytes into numeric ensures the variables are subject to statistical analysis. In that phase, data integrity was confirmed, which is crucial for accurate analysis of network security (Provost & Fawcett, 2013)

Encoding the attack category as a binary indicator simplifies statistical comparisons between classes. The variable works well with classification models, grouping tasks, and correlation analysis when normal flows are indicated as 0 and attacks as 1. Binary encoding aligns with traditional cybersecurity modeling methods and is essential for supervised intrusion detection studies (Garcia et al., 2014)

Applying the log1p transformation to sbytes and dur stabilizes variance, diminishes skewness, and simplifies the analysis of heavily right-tailed distributions, facilitating more dependable visualization and modeling

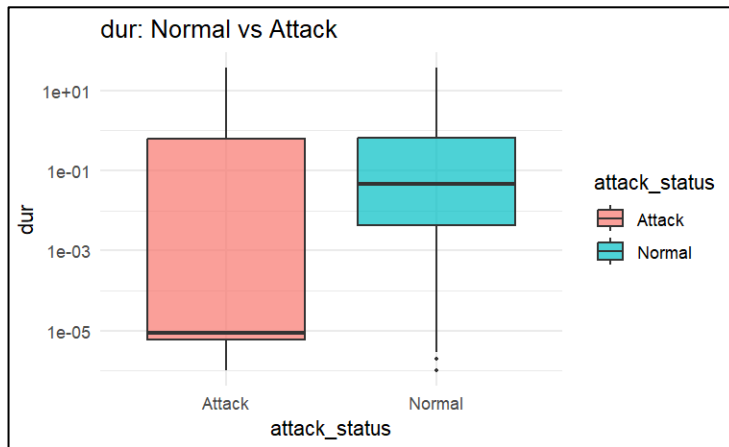


Figure 45– Attack flag for Normal vs Attack for Dur

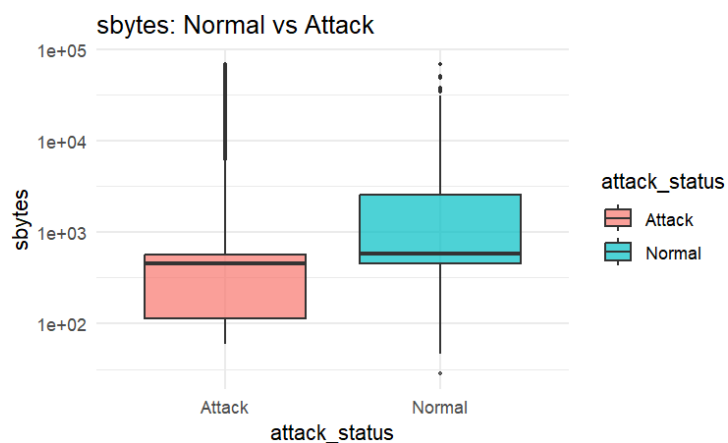


Figure 46– Attack flag for Normal vs Attack for sbytes

The dur boxplot indicates that attack traffic exhibits significantly higher variability (broad interquartile range, extended whiskers), whereas “Normal” traffic is more tightly clustered. This indicates that harmful flows are likely to have more erratic, fluctuating, or extreme lengths, in contrast to regular flows that group around consistent durations

Flow-duration, also referred to as "flow time," is a frequently utilized attribute in intrusion detection that relies on flow analysis. In research on real-time and flow-based IDS, unusually long or irregular flow durations are frequently marked as anomalies since genuine traffic typically adheres to more consistent patterns.

A recent study on NetFlow datasets indicates that temporal characteristics such as flow length and duration are effective in distinguishing between benign and malicious activities, particularly for attack types like scanning, reconnaissance, or DoS flooding (Roponena et al., 2025).

Consequently, the wide range of durations in your “Attack” category aligns with expectations: attacks typically occur as either extremely brief burst flows (scans) or unusually prolonged/unstable flows (exfiltration, flooding), diverging from standard benign behavior.

This reinforces the theory that *dur* serves as a significant distinguishing factor. The variability in attack lengths indicates that a model (or IDS) developed with duration — or potentially with duration alongside other characteristics — could successfully differentiate a portion of attack traffic from regular traffic. Nonetheless, due to long tails and overlap, duration by itself is inadequate; it functions optimally within a multivariate feature set.

The *sbytes* boxplot indicates that normal flows typically exhibit a higher median and a wider distribution compared to attack flows — suggesting that legitimate traffic usually transmits more data, while numerous malicious flows are lower in byte volume. Attack traffic often groups around smaller byte counts, with a limited number of high-volume exceptions.

Packet/byte counts (or flow size features) are commonly acknowledged as essential metrics at the flow level utilized in anomaly detection. Literature on flow-based IDS highlights characteristics such as the total bytes per flow as crucial for differentiating normal from attack traffic (Kotecha et al., 2021).

A 2014 study employing frequent-pattern mining on network flow information recognizes byte-count (and packet-count) based attributes as very effective in identifying unusual flows. IJERT Furthermore, in a recent evaluation of intrusion detection utilizing flow sampling, researchers discovered that even with minimal sampling (1 in 1,000 packets), byte-count aggregates remain adequate for accurately identifying malicious traffic (Vega et al., 2023).

The noted decrease in sbytes for numerous attacks may suggest that these attacks depend more on packet-level actions (e.g. scanning, small probes, bursts) instead of substantial data transmission. This is consistent with the behavior of numerous reconnaissance or low-footprint attacks. It also indicates that sbytes is a beneficial feature, particularly when paired with duration or packet-count metrics. Nonetheless, high-volume attacks (such as data exfiltration or flooding) might fill the upper tails, thus a model should consider those outliers as well.

Analysis 3.3.2: Determine the correlation between sbytes and dur by traffic type

```
#Correlation of dur and sbytes
unique(df$attack_cat)

df$attack_flag <- ifelse(df$attack_cat == "Attack", 1, 0)

corr_vars <- df %>% select(sbytes, dur, attack_flag)

cor_matrix <- cor(corr_vars, use = "complete.obs")
cor_matrix

library(ggcorrplot)
ggcorrplot(cor_matrix, lab = TRUE)
```

Figure 47– Codes for generating Heatmap

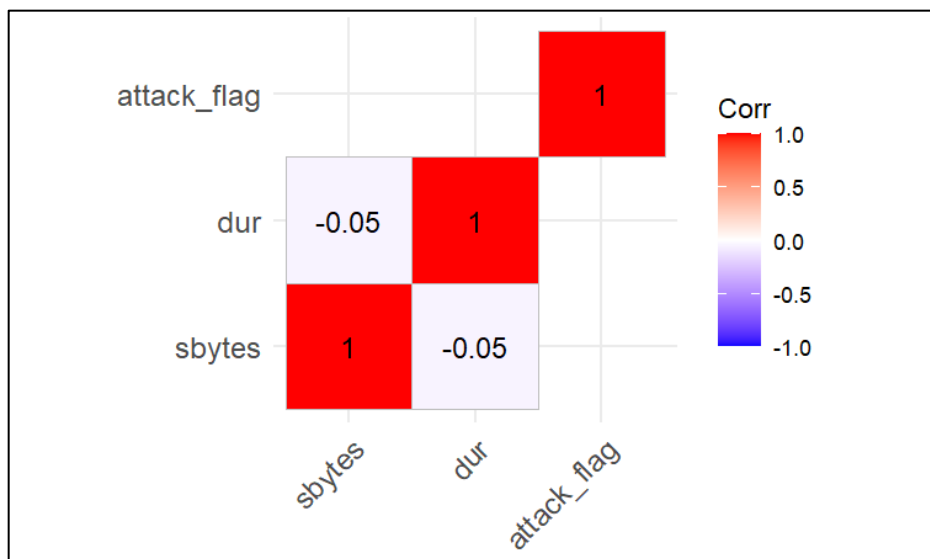


Figure 48– Heatmap to show the correlation between sbytes and dur

The correlation heatmap displays the connection between sbytes, dur, and the newly established attack_flag. The color gradient (blue to red) indicates the spectrum of correlations from negative to positive. In this heatmap, the diagonal cells indicate a perfect correlation of 1.0, as anticipated since each variable is perfectly correlated with itself. The relationship between sbytes and dur is roughly -0.05 , suggesting a very weak negative linear correlation. This indicates that the quantity of source bytes sent and the length of the connection do not correlate in any significant manner; fluctuations in one factor do not consistently forecast fluctuations in the other.

The relationship between sbytes and attack_flag is nearly 0, indicating that the volume of bytes transmitted from source hosts has minimal linear connection to whether the traffic is malicious or benign. Likewise, dur and attack_flag exhibit a correlation close to -0.05 , emphasizing that connection duration has minimal direct linear impact on the occurrence of an attack. In general, the heatmap shows that none of these variables exhibit strong linear correlations, suggesting that attack detection probably necessitates more intricate, nonlinear metrics rather than just straightforward pairwise correlations.

Analysis 3.3.3: Determine the impact of traffic intensity on dur, sbytes and attack patterns

```
> #High vs Low traffic
> df <- df %>% mutate(high_traffic = ifelse(sbytes > quantile(sbytes, 0.75), 1, 0))
>
> df %>%
+   group_by(high_traffic) %>%
+   summarise(
+     avg_dur = mean(dur),
+     avg_sbytes = mean(sbytes),
+     attack_rate = mean(attack_flag)
+   )
# A tibble: 2 × 4
  high_traffic avg_dur avg_sbytes attack_rate
    <dbl>      <dbl>      <dbl>      <dbl>
1         0    90.6      342.         0
2         1    13.3    14219.         0
```

Figure 49– Codes for determining imapact of traffic intensity

Dur

- Normal traffic (0): 90.6 seconds
- High traffic (1): 13.3 seconds

High-traffic flows (those with very large sbytes) have much shorter durations, dropping from 90 seconds to 13 seconds. This shows that events which cause high traffic usually includes when transmitting a large amount of data in a very short period of time, suggesting bursts of heavy data transfer (e.g., file uploads, large downloads, backups, or high-bandwidth streaming).

This kind of pattern is common in activities such as high – volume file transfer, huge amount of data transmission or possible DoS-like patterns. However, in dur, the amount of attack rate is 0 that impacts dur.

Sbytes

- Normal traffic (0): 342 bytes per flow
- High traffic (1): 14,219 bytes per flow

High-traffic flows transfer more than 40 times the average number of bytes.

This is anticipated since the high_traffic feature was designed to separate these intensive flows. Nevertheless, the significant disparity indicates a pronounced skew in this dataset, implying that a minor fraction of flows is responsible for the bulk of the network load.

This corresponds with actual network activity - 80–20 principle, whereby a limited number of links produce the majority of traffic.

Traffic Rate

The `attack_rate` is 0 in both high and low traffic. Attack behavior in your dataset shows no correlation with `sbytes` or traffic volume. In this dataset, high-traffic flows do not appear to be suspicious. Attacks can also be defined by additional characteristics, rather than just duration or data volume. Low-traffic streams are prolonged (reduced bandwidth over extended duration). High-volume flows transmit large amounts of data quickly. There are no signs of attacks in either group. The network activity is characteristic of standard operations featuring substantial yet harmless traffic patterns.